Foundations of causal inference (cont.)
Experimental Designs

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1 Online ad experiments in marketing and their problems

Recall

- **Intent to treat (ITT)**
  - Under non-compliance, where $D(T)$ does not always equal T
  - ITT is a valid estimate of the effect of the assignment T, but <u>not</u> the effect of treatment D
- **Local average treatment effect (LATE)**
  - Average treatment effect of the compliers
- $\text{ITT} = \text{ITT}_D \times \text{LATE} = p(\text{complier}) \times \text{LATE}$

1.1 Relation between ITT and ATET/ATE

- ITT is often seen as a conservative estimate of ATE / ATET
  - ITT is typically smaller than LATE under monotonicity (no defiers)
  - $\text{ITT} = \text{ITT}_D \times \text{LATE}$
  - $\text{ITT}_D = \mathbb{E}[D_i = 1|T_i = 1] - \mathbb{E}[D_i = 1|T_i = 0]$ is between 0 and 1 $\rightarrow$ share of compliers
- However, this relation is only valid when both the instrument (here, assignment T) and the focal treatment variable (here, exposure D) are binary: $T, D \in \{0, 1\}$
  - Otherwise, $\mathbb{E}[D_i = 1|T_i = 1] - \mathbb{E}[D_i = 1|T_i = 0]$ also does not need to be in the 0 to 1 range
  - We will look at that situation again when discussing instrumental variables in more detail

1.2 When is LATE = ATET?

- One-sided non-compliance in the treatment group
  - When the treatment is only available to the treatment group, the compliers are exactly the same group as the treatment population (ATET)
  - We only have compliers and never-takers, or $D_i(T_i) = 1$ only for $T_i = 1$
- Perfect compliance
  - If every individual complies with their assignment, i.e., $p(\text{complier}) = 100\%$ or $D_i(T_i) = T_i$
- Homogenous effects
  - If the effect of compliers and always-takers are the same, then LATE = ATET
  - If the effect of compliers and all others are the same, then LATE = ATE
    - i.e., there is **no treatment effect heterogeneity**

A typical question in (digital) marketing

- Is it worth spending money on (digital) advertising?
- Is there an incremental effect on (digital) advertising?
  - What would have happened if the advertising was not present?
  - The difference in the outcome in the absence and presence of the focal advertisement (treatment)
- Again, advertising exposure is not random. People are explicitly targeted for advertising where they show interest in products

  - We need a randomized experiment (or full information about all targeting variables → usually impossible)

- Many ad platforms offer tools for ad experiments; i.e., randomized A/B tests

1.3 Single-ad study vs. multi-ad studies and causal inference / divergent delivery

1.3.1 The problem with divergent delivery

- In online experiments, there is often a difference between users assigned to an experimental group adn those that are actually exposed to the ad
  - Non-compliance
- Randomization is based on the user-level
  - All users who are eligible to see the ad
- But not all users in the treatment group will be exposed to the ad
  - Exposure is decided on an incidence-level (each time an ad is loaded)
- The targeting and real-time auctioning process exposes users to the ads depending on their consumer profile, a prediction of whether they are likely to respond, and competitors' bissing behavior
  - Consequently, the exposed users are not a random sample of eligible users
    * Divergent delivery // non-random compliance

1.3.2 Two main types of online ad experiments — divergent delivery

Single-ad study with holdout/control

- Typical question: is there an incremental effect of the ad on an outcome of interest?
- Consumers are randomly divided into treatment group and a holdout/control group
- Consumers in the treatment group are eligible to see the ad, but not all consumers will see the ad
- Consumers in the control group will not see the ad
  - One-sided compliance, we are able to calculate LATE or even ATET in some situations

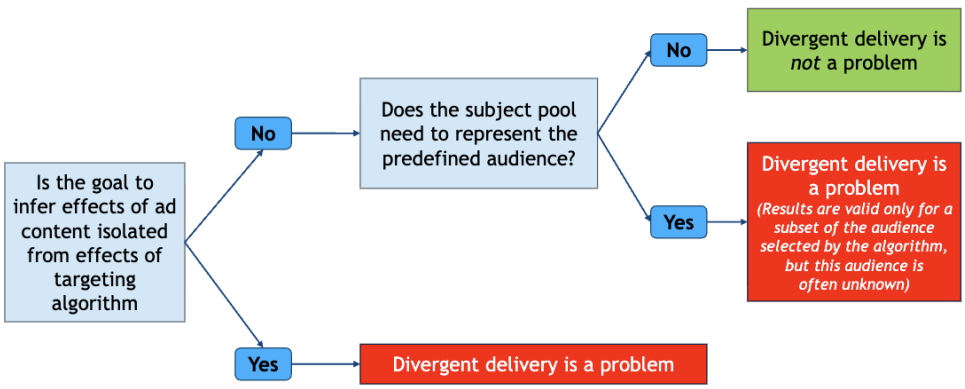
Multiuple-ad study without holdout

- Typical question: Which of the two versions of an ad is better?
- Classic A/B testing: Consumers are randomly assigned to group A or B
- Consumers in group A will only see ad A
- Not all consumers will see the ad because of targeting (exposure is not random, only assignment)
  - We are not able to make inferences about incremental effects of A over B; i.e., about causal effects

1.4 Interpretation of A/B test (multiple-ad study) with targeting

- Under divergent delivery, results may be interpreted as the combined effect of ad content and platform's targeting
  - Whether this matters to the experimenter depends on the goal of the experiment
- You cannot say that ad A is better than ad B (in general or for the specified audience)
- You can only say that ad A is better on this platform given the specific targeting algorithm and campaign settings
- Ad B might be similar or better than A on a different platform, with a different algorithm, with different optimization objective

1.5 When is divergent validity a problem for multi-ad studies?



1.6 Single-ad study with holdout/control

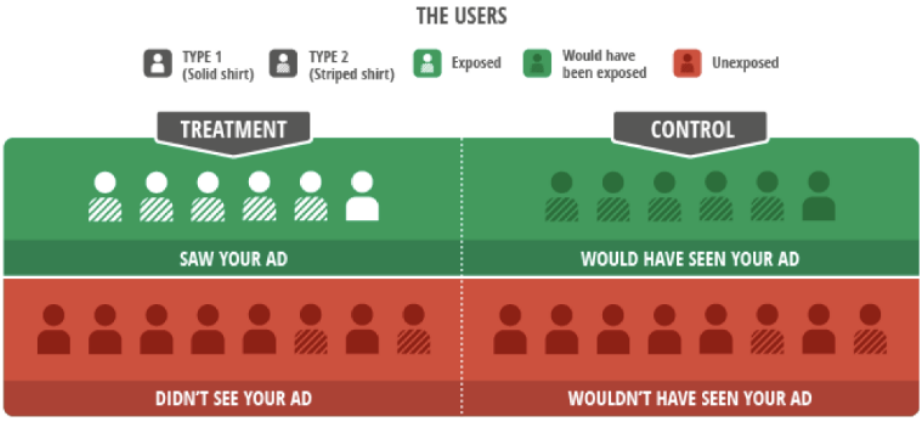
- Typically, we have one-sides non-compliance (users in the control group do not see the ads)
- We may be able to estimate ITT and LATE, but typically they will not be a good estimate of ATE or ATET, because compliance is not random and there may be treatment effect heterogeneity

1.7 Identification of outcomes

- How to track individual outcomes (e.g., website visits, registrations, purchases) in the non-treatment group (i.e., those users that will not see the ad is often challenging)
  - The tracking cookie is ususally placed with the ad
  - We need a platform that identifies control user outcomes
  - We may need to be able to differentiate between control users that would have been exposed
- Not all marketing relevant metrics can be used as outcomes
  - All metrics that require seeing an ad are not possible (e.g., click-through rate: CTR)

1.8 The ideal situation — ATET

- Ad effectiveness as the difference between customers who saw the ad and consumers who would have seen the ad, had they not been in the control group



1.9 Conclusions

- Significant selection effects usually lead to overestimating the ad effect
- Observational methods are usually not able to estimate the ad effect well enough
  - Even with excessive control variables they overestimate in most of the studies
  - No single observational method is systematically better
- Randomized experiments (AB testing) is the gold standard to avoid selection effects
- Individuals usually vary in their strength of the treatment effect
  - Instead of looking only at averages we can also try to explore heterogeneity in the treatment effects → estimate a CATE
    - e.g., group specific estimates, interaction effects
- Individuals might not always comply with treatment assignment
  - One-sided or two-sided non-compliance
  - Compliers, always-takers, never-takers, defiers
  - Treatment effect estimators: ITT, ATET, and LATE
- For measuring (online) advertising effectiveness we need properly designed experiments
  - We need understanding about one-sided non-compliance etc.
  - Even excessive amounts of control variables are not able to properly control for all selection effects

1.10 A bit on binary outcomes and causal inference

- In marketing, we often have binary outcomes (choices, events, etc.)
  - Responded (yes/no), purchased (yes/no), choice (A/B)
  - Marketers are often interested in conversion rates

Group	Purchased (1)	Did not purchase (0)	Total
Ad	18	82	100
No ad	10	90	100

- Three (four) different focal measures
  - Risk difference
  - Risk ratio (or lift)
  - Odds ratio
- **Risk difference**:
  - 18% - 10% = 8% $\rightarrow$ our ATE
  - The ad increases the probability of purchasing by 8 pct
- **Risk ratio:**
  - Compares the probability of an event happening in one group vs. another
  - $\frac{18}{10} = 1.8\%$
  - People taking seeing the ad are 1.8 times as likely to purchase compared to those not seeing ads
- **Odds ratio:**
  - Compares the odds of an event, not the probability; i.e., the ratio of successes to failures
  - Odds in treatment: $\frac{18}{82} \approx 0.22$
  - Odds in control: $\frac{10}{90} \approx 0.11 \rightarrow$ odds ratio: 1.98 (odds in treatment over odds in control)
  - The odds of purchasing are 2.68 times higher when seeing the ad vs not seeing the ad
- In marketing A/B testing, lift is often used as a relative measure of improvement in a rate (conversion, click-through, retention)
- The risk-ratio (RR) is almost the same as "lift", just expressed as a ratio rather than a percent:

$$\text{RR} = \frac{\mathbb{E}[Y=1|T=1]}{\mathbb{E}[Y=1|T=0]}$$

$$\text{Lift} = \frac{\mathbb{E}[Y=1|T=1] - \mathbb{E}[Y=1|T=0]}{\mathbb{E}[Y=1|T=0]} = 1 - \text{RR} = \frac{\text{Risk difference}}{\mathbb{E}[Y=1|T=0]} = \frac{\text{ATE}}{\text{Baseline conversion}}$$

Previous example

- Lift $= \frac{0.18-0.1}{0.1} = 1 - 1.8 = 80\%$
- Ads improve conversion rate by 80%

1.11 Logistic regression

- Using logistic regression seems natural when having binary outcomes
- However,
  - Logistic regression is typically not estimating ATE (the risk difference) nor the risk ratio (lift) which is what decision-makers are typically interested in
  - It estimates the treatment effect on the odds-ratio (often not what decision-makers are interested in)
- Logistic regression is superior to linear probability models (linear regression on binary outcomes) when it comes to predictions, but not necessarily causal inference (*depending on the type of question*)

2 Experimental Design

2.1 Why it matters

- Experimental design is the planning stage of an experiment
- The task of the experiment is to get an answer to a business/research problem
  - The task of experiment design is getting the right data so you can get an answer that's valid
- From: what is it that we want to study? ... via... how do we implement the experiment in a way that it creates valid conclusions to our questions? ... to... analyzing the data in a way that reflects the experimental design / actual implementation
  - Usually, the planning makes more than 80% of the time and resources, while less than 20% is data analysis
- Without careful planning of your experiment, you are wasting resources because even the cleverest data might not recover the core question
  - Ensuring that statistical tests / data analyses are able to answer the question we want to solve

2.2 Within vs. between subject experiments

- **Between subjects**
  - Treatment and control measure are taken from **different experimental groups** with different test subjects
  - Together with **random assignment** it is most effective in isolating the treatment effect
- **Within subjects**
  - Treatment and control measure are taken from the same test subjects at **different points in time**
  - Need to assure that **nothing changes** between the two time points beside the treatment
  - Prone to several biases
    - Demand effects
    - Order and carryover effects
    - Changes in the environment

2.3 Manipulating the independent variable

- A core challenge in the design of every experiment is to find the right way of **varying the treatment** (i.e., the **independent variable**)
  - Selecting the right treatment and control condition(s)
- What is the **counterfactual?**
  - Is it the absence of something?
  - Or is it an explicit alternative?
  - What is the focal concept that we want to test / research?
  - How can we correctly implement the alternatives without inference?
- Balance between realism and a clear definition of the desired manipulation
  - Internal vs. external validity
- Variation of the independent variable must meet two important requirements:
  - Effectiveness: The treatment must be effective in the sense that the focal concept is indeed varied
  - Control: The treatment must guarantee that any spects other than the focal independent variable are not unintentionally varied

2.4 Examples

- **Scarcity queues in E-commerce**
  - Causal question: does displaying a scarcity queue (few products left in stock) influence purchase intentions / sales in e-commerce stores?
- **IV**: scarcity
  - Treatment: displaying in-stock information
  - Counterfactual / control: ???
- **Effectiveness**: is displaying in-stock information actually varying scarcity perception?
  - Is the displayed number large/small enough to affect scarcity in a meaningful way?
  - Manipulation check: measure scarcity?
- **Control**: is displaying in-stock information varying any other variables than scarcity?
  - e.g., it could signal popularity, etc.

2.4.1 Compound treatment — example

- You want to investigate the causal effect of a price discount on purchase decisions
- You run a typical A/B test
  - You randomly assign customers to two groups
  - Customers in the treatment group get an email with the price discount
  - Customers in the control group do not get an email
- **Is the difference between the groups the effect of the price discount?**
  - I'd say no — the effect is most likely from spamming customers with emails. To get the true effect we must simply do a price change on the website with no other actions

2.4.2 Similar: Placebo effects

- You want to test the effect of receiving a medicine on reduction of pain
- You run a typical A/B test
  - You randomly assign customers to two groups
  - People in the treatment group get the pill
  - People in the control group do not get the pill
- **Is the difference between groups the effect of the medicine?**
  - No, because observed differences between the treatment and control group may be due to *receiving the pill* and not because the medicine actually works
    - Solution: also give a placebo pill to the control group

2.5 Measuring the dependent variable(s)

- Another core challenge is to find a good way to measure the dependent variable(s); i.e., the intended outcome variable(s)
- What is the variable that we are interested in?
  - Are we able to measure it directly? Do we need proxy variables to measure it indirectly?
  - Are we able to measure it in all treatment (and control) conditions the same way?
    - Is measuremetn possible in the absence of treatment?
- Does the type of measure introduce threats to validity?
  - e.g., demand effects, anchoring and carry-over effects, social desirability, etc.?
- Do we need to capture additional variables to understand the process and mechanism of the treatment better (i.e., mediators)?

2.6 Validity of experiments

- **Internal validity**
  - Refers to whether the treatment of actually caused the observed effects o the dependent variable(s)
  - Can we actually estimate the treatment effect within the given context and for our particular sample?
    - Fails when there are differences between treatment groups (other than the treatment itself) that affect the outcome and that we cannot control
    - Directly associated with the design of the experiment
- **External validity**
  - Refers to whether the cause-and-effect relationships found in the experiment can be generalized
  - Can we extrapolate our estimates to other settings / pupulations / time?
    - Fails when the treatment effect is different
      - outside the evaluation environment
      - or for other groups of people
    - How realistic / representative is the experiment for real environments in which the treatment should be applied?

2.7 Internal validity

2.7.1 Some common threats to internal validity of experiments

- Statistical conclusion validity
- The experimental protocol
  - Potential outcomes not independent of assignment: presence of selection effects
  - Non-compliance
  - Interference / violation of SUTVA
  - (Differntial) attrition
  - Time effects
- Awareness about the experiment
  - Hawthorne effect
  - Demand effects
- The measurement of the outcome
  - Differential measures
  - Ambiguous temporal precedence

2.7.2 Threats to internal validity — statistical conclusion validity

- Are the statistical assumptions of the method in use fulfilled?
  - Affects our ability to accurately estimate the treatment effect and correct inference from data
  - Some potential problems:
    * Misspecification of the functional term (e.g., linear, non-linear)
    * Misspecification of DV: continuous vs. discrete
    * Misspecification of the error structure: e.g., heteroskedasicity or autocorrelated error terms
    * Ignoring measurement error
- Knowledge of the design is important to choose the right statistical framework for estimating the causal effects under given circumstances
  - Different designs require different statistical methods
  - Some designs may never allow correct statistical inference

2.7.3 Threats to internal validity — The experimental protocol

- Failure of randomzation / selection effects
  - If the assignment mechanism is not random, we cannot rule out selection effects
  - We can test on observed pre-treatment variables for equivalence of groups
  - Or we need to require conditional independence (i.e., after controlling for observables we achieve independence)
- Non-compliance
  - Subjects do not comply with their treatment status
  - e.g., compliers, always-takers, never-takers, defiers
- Violation of SUTVA (stable unit treatment value assumption)
  - Interference: potential outcomes are not independent of the treatment status of other individuals

2.8 Stratified random assignment

- Does not randomize on the population level, but within more homogenous subgroups
  - Assigns loyalty card members with 50% probability to treatment and control
- Endures better balance for each of the characteristics on which we build subgroups/strata
  - Characteristics should be potentially confounders where balancing is important
- Important if the characteristics have low prevalence in the population or we have small sample sizes
  - Example: we have a sample of 100 customers of an airline out of which 2 are heavy frequent flyers. We split the 100 customers into two randomly assigned groups. What is the probability that the frequent flyers end up in one group?
- Important if we want to use this characteristic to analyze heterogeneous treatment effects
  - e.g., sub-group specific ATEs

2.9 Interference problems in marketing — examples

- **Word-of-mouth**
  - Treated individuals may affect the outcomes for untreated individuals by talking about the treatment
- **Observation**
  - Untreated individuals may observe purchasing behavior of treated individuals and adapt their purchase behavior
- **Crossover**
  - Individuals who are assigned to the control are (accidentally) also exposed to the treatment
- **Network effects**
  - Whether other individuals in the network are treated or untreated increases/decreases the consumption utility (i.e., changes the potential outcomes)
- Treatment effects are not independent of the treatment status of other individuals

2.10 Threats to internal validity — the experimental protocol

Differential attrition

- Attrition: Participants leave the experiment before all measurements are completed
- Differential attrition: Drop out in the treatment group is larger or smaller than in the control group due to reasons of the experiment / drop out is not independent of the potential outcomes
  - e.g., treatment or no treatment causes some people to drop out

Time effects

- When treatments are administered at two different times, outside events, learning, or other changes in the environment can influence the treatment effect

2.11 Threats to internal validity — awareness about the experiment

Hawthorne effect

- If individuals know that they are part of an experiment, they may change their behavior due to the attention and monitoring they receive

Demand effects

- Participants guess the purpose or goals if the experiment and try to conform (or disconform) with it
- They change their behavior in favor for or against the hypothesis → increasing or decreasing the treatment effect

Ideally people should be **unaware of the experiment** but consider ethical consquences

2.12 Threats to internal validity — the measurement of the outcome

Differential measures

- Sometimes it is difficult to measure the same outcome (or with the same reliability) in the different treatment groups
- Psychological measures may change meaning due to treatment (or absence thereof)
- Actual behavior might be harder to observe in the absence or presence of the treatment
  - e.g., tracking cookies that come with an advertisement, but is not placed without the ad
  - e.g., engagement (e.g., clicks) is not possible if not seeing an ad

Ambiguous temporal precedence

- The type of measurement does not allow to distinguish whether treatment actually lies before the outcome or after → resolution of the measurement is too broad (e.g., day, week, month)